

HW 5: Errors and Power

1) A TRS agent has to make a decision as you go through the airport security line. They assume all passengers going through do not have a prohibited item, but they are trying to find evidence that you do. Write down the null and alternative hypotheses in words, complete the table, then answer the questions that follow.

H_0 :	Truth about the Population		
		H_0	H_a
H_a :		Truth:	Truth:
a) Describe a Type I error in context.	H_a	Decision:	Decision:
b) Describe a Type II error in context.	H_0	Truth:	Truth:
		Decision:	Decision:

2) For the following null and alternative hypotheses, determine if the statements are true (T) or false (F).

	_____ a) A Type I Error occurs when we incorrectly conclude that a person has the disease when they do not.
	_____ b) The power of this test would be the probability of correctly diagnosing a person with the disease when they do have it.
H_0 : A person does not have a disease	_____ c) Decreasing the sample size in a test aimed at detecting the presence of a disease would increase the power of the test.
H_a : A person does have a disease	_____ d) If the null hypothesis states that a person does not have a disease, rejecting the null hypothesis implies that the person definitely has the disease.
	_____ e) The probability we observe data that the person has the disease, assuming they do not actually have the disease, is the p-value.

3) Given the following statements, determine if they are always, sometimes, or never true.

- _____ a) When all other aspects of a test remain the same, increasing the sample size will reduce the probability of committing a Type II Error.
- _____ b) The power of a test will increase as the sample size increases and your sample proportion gets closer to 0.5.
- _____ c) When all other aspects of a test remain the same, increasing the significance level increases the chance of committing a Type II Error.
- _____ d) A smaller p-value indicates stronger evidence against the null hypothesis.
- _____ e) Increasing the sample size will always decrease both Type I and Type II Error rates simultaneously.
- _____ f) Lowering the significance level will always decrease the probability of committing a Type I Error, regardless of sample size.
- _____ g) A Type II Error occurs when the null hypothesis is true.
- _____ h) The power of a test is always equal to 1 minus the probability of a Type II error.
- _____ i) If the significance level of a test is set to 0.01, the probability of committing a Type I Error is higher compared to when α is set to 0.05.
- _____ j) A Type I Error can occur even when the null hypothesis is true.

4) The administration of a university wants to improve student satisfaction with the registration process. Historically, it is known that 70% of students experience wait times of more than 15 minutes during registration. To address this issue, the administration implements a new online registration system they believe will reduce wait times. They plan to conduct a study to determine if the proportion of students experiencing wait times over 15 minutes has decreased after the implementation of the new system. They will randomly select a sample of students during the registration period and compare the proportion of students experiencing long wait times before and after the implementation of the new system.

- a) What is the parameter of interest in this problem? b) State the null and alternative hypotheses.

c) Describe a Type I and Type II error in this setting and explain the consequences of each.

d) The higher the statistical power of a test, the lower the risk of making a Type II error. Power is usually set at 80%. This means that if there are true effects to be found in 100 different studies with 80% power, only 80 out of 100 statistical tests will actually detect them. What are two ways that we can increase the power of a test?